KOÇ ÜNİVERSİTESİ
COMPUTER ENGINEERING

COMP132: Advanced Programming
Programming Project Report

<NBA SIMULATION>

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Application Usage

First, you need to register for the game by clicking the create account button then you need to log in to play this beautiful game. After that, you will pick your team, and the NBA draft going to open. You can pick your favorite players if they haven't been picked by other teams yet. Then you should click the continue button. Now you can simulate the in-season matches and see the result of those matches on the screen if your team is in the best 8 teams in the simulation your team will be eligible to play playoffs then you can click the Continue Playoffs button and simulate the Playoffs as well at the end you can see who the champion of the game is! Have fun!

Introduction

I was hesitant to start this project at first, but then I was given a chance to learn some GUI because I am very interested in the visible elements of the computer. Then I learned how to create a label, panel, combo boxes, etc. in GUI. I couldn't use Windows Builder due to some problems with my compiler, so I placed everything you see in my project GUI manually. It was challenging at first, but I got used to it after I experienced what everything does in GUI. In my project report, I would like to give you a precise explanation of all my classes and packages.

1-NBAGAME.java

Let's dive into this class, This is where I planted the first seeds of my project. Beginning of everything you see. NBAGAME.java is inside the myGui package which is the package that contains the classes that have their panels.

I wrote the GUI code for login here and I added the images that I thought fit this project nicely. We have createAccountButton and changeAccountInfoButton those are buttons that open different panels for different purposes. And we have a "PLAY" button to let us dive into the game. Then I have a method for checking user has already created his/her account or not.

I'm reading the text file called users.txt and since it is a Boolean method, I'm checking whether the user is valid or not by returning true or false. Then as you can see, I have a method called playBackGroundMusic its purpose is just to make this game more beautiful, I'm adding a WAV music file to my program to increase our users' joy. The rest of the class methods are implemented to follow the purpose of my other classes.

2-PickTeam.java

When we pushed the PLAY button in NBAGAME.java we dive into PickTeam.java. I passed the panel and label with parameters. Here I added all real NBA team logos, which I used for the loop because all my image paths were similar for example path1=IMG_1_PNG, path2 =IMG_2_PNG… so I used that in for loop and added all my images to my panel that way. I wanted to make all images a button as well, that was challenging stuff to figure out. Because I needed to modify the pictures and the only purpose of the class called CustomImageButton.java was that one. After the user picks its logo, the button lets us dive into DraftPage.java.

3-DraftPage.java

The purpose of this class is to let our user to pick his/her team's players. I implemented the images of most known players in basketball positions. Then my draft mechanism backend appears. (You need to know the information of the ComputerBot.java. please read the text below first.) My approach was to add the user itself named "User" as a computer bot and 29 computer bots named "1","2","3" …"29". I added all including the User inside a computerBots array list which takes computerBots. Then I shuffled this list. After that you should see that I used a method called readPlayerPositions its purpose is to read the given files and add the players to their correct combo box and array list (by their position). After that, I used a for loop for every computer bot and User itself to draft their players. If the computerBotsList's index doesn't contain the element "User", the computerDraft() method inside the NBAGAME.java will work and automatically our computer bot will have its players. But If the corresponding index is equal to “User” then our combo boxes will enable us to give us their players. Right after that, I used showTeam.java’s show() method in the class the purpose of this method is to let us see our team members and their specific specialties. After the getters and setters you should see where I store the information of the draft results in draft_results.txt I used bufferwriter and filewriter for that purpose.

4-ComputerBot.java

This class represents a computer Bot. At first, I created the array list that will represent the players by their position and a picked list for ComputerBot’s drafted players. Then you can see my constructor and a method that I’m determining the computer team’s logo. Rest is just getters and setters.

5-Player.java

The purpose of this class is to read the given CSV file and store the players in 5 different Txt files by their position. I implemented different approaches for TOT’s and others. You can see it in my while loop. I added all players with TOTs inside a set and checked whether the player had TOT or not that way. I differentiate the player classes in the package called PlayerTypes.

6- PointGuards.java, ShootingGuards.java, SmallForwards.java, PowerForwards.java, Centers.java

The purpose of those classes is just to determine the overall of the player, or you may say the power of the player. They read their specific text files and determine the power of the player (I used my basketball knowledge to determine that one.) and store that information in new txt files called powerOf+position+.txt.

7- Matches package classes (Matchmaking.java, Matchscreen.java, SimulationControl.java

This package is meant to control, specify, determine, and GUI elements of matchmaking. I used three different classes for different approaches.

Firstly, I needed to determine how the Match process works and how can I implement those specifications in my code briefly. I created the class called MatchMaking.java for that purpose. I have several methods in this class, determineGameWinner method is basically to determine the winner of the game, it takes two different computer bot teams as parameters since the user itself is a computer bot named “User” this approach is valid. I’m changing the team stats due to the winner of the game. And since in real basketball, there is no such thing as a tie in the games if that situation appears I’m choosing the winner randomly and incrementing the winner's score and changing the team stats too. At the very end you should see the playoffMatchDetermineWinnerTEAM method it has similar purposes to determineGameWinner I used different methods for playoff and in-season because playoff match results must be stored in different text files. (I do that in the method called writeGameResultToFile at the end of this class)

playRandomGames method is for determining who plays with whom. I used a random generator for that purpose.

updateTeamStats method is for increasing the number of wins of the winner and decreasing the number of losses of the loser.

readPlayerPower method, this method is for reading the txt files that start powerOf… I used a buffer reader and file reader and checked the second value of the txt files since the first value is the name of the players.

calculateTeamScore method, this method is for determining the power of the team. I increment the variable called totalPower with all the player powers. And then I added randomness to it to make this game more realistic.

MatchScreen.java contains most of the GUI elements of the matches. Images, buttons, etc. In this class, I wanted to make an animation for loading to make this game more beautiful. I used the Timer class of the java for that purpose. At the end of the class, you should see the teamSpecialities button, this button is for us to see our team members and their specialties, the code takes us to the showTeam class’s show method. I have another method called updateMatchScreen, here I’m placing the best 8 teams in the tournament based on their number of wins and I used Java’s Component class to erase the labels because I’m implementing this method inside a loop in SimulationControl.java so I needed to erase the labels that I implemented before due to prevent large number of images.

Our final class in the Matches package is SimulationControl.java let me explain what I have done in this class, The SimulatePlayoffSeason method contains GUI elements of the NBA playoffs. I made an array list called logopaths outside the method for the top 8 team logos and I placed those logos into the panel according to the given playoff tree. Then I have two buttons, The team specialties are the same as the previous classes and I have simulated the playoff button is simulating the playoff with using the methods inside the Matchmaking.java and I placed the logo label of the winning team in the playoff tree until we have a champion!

simulateInSeason method, firstly I'm adding GUI elements such as JLabel and then we will see a while loop I determined how many updates that I needed (I picked the number of 6 due to the real NBA in-season results). And I implemented two more methods for pausing and resuming the simulation but unfortunately, it doesn't work quite well.

8- CreateAccount package

Inside the CreateAccountPage.java I implemented the GUI elements of the page and some validations for correct user input.

In ChangeAccountInfo1.java I let the user log in first and in another class called ChangeAccountInfo2.java I add the visual elements it such as labels.

Reference:

[1] *javax.swing (Java Platform SE 7)*. (n.d.). javax.swing (Java Platform SE 7). Accessed: 25 Dec,2023, https://docs.oracle.com/javase/7/docs/api/javax/swing/package-summary.html